Complex number

If $z = x+iy$, then, $x \rightarrow$ real part of z.

$y \rightarrow$ imaginary part of z.

Complex numbers are the numbers that can be expressed as $a+ib$, where a is the real part and b is the imaginary part.

Complex conjugate, $\bar{z} = x-iy$.

Addition,

$$(a+ib)+(c+id) = (a+c)+(ib+id) = (a+c)+i(b+d)$$

Substraction,

$$(a+ib)-(c+id) = (a-c)+i(b-d)$$

Multiplication,

$$(a+ib)*(c+id) = ac + iad + ibc + bdi^2$$

$$= (ac-bd)+i(ad+bc)$$

Division,

$$(a+ib)/(c+id) = \frac{a+ib}{c+id} \cdot \frac{c-id}{c-id} = \frac{ac-iad+ibc+bd}{c^2+d^2}$$

$$= \frac{(ac+bd)+i(bc-ad)}{c^2+d^2}$$

$$= \frac{ac+bd}{c^2+d^2} + i\,\frac{bc-ad}{c^2+d^2}$$

Absolute value or modulus –

$$|z| = |a+ib| = \sqrt{a^2+b^2} = r$$

Polar form,

$$x = r\cos\theta \quad ; \quad y = r\sin\theta$$

$$\therefore z = x+iy = r(\cos\theta + i\sin\theta)$$

De Moivre's theorem –

Let, $z_1 = x_1 + iy_1 = r_1(\cos\theta_1 + i\sin\theta_1)$

$$z_2 = x_2 + iy_2 = r_2(\cos\theta_2 + i\sin\theta_2)$$

So, $z_1 z_2 = r_1 r_2 \{\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)\}$

$$z_1/z_2 = r_1/r_2 \{\cos(\theta_1-\theta_2) + i\sin(\theta_1-\theta_2)\}$$

or, $z^n = r^n(\cos n\theta + i\sin n\theta).$

This is called De Moivre's theorem.

* * *

Euler's formula –

By assuming that the infinite series expansion –

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \text{- - - -}$$ of elementary calculus holds when $x = i\theta$, we can arrive at the result

$$e^{i\theta} = \cos\theta + i\sin\theta$$

This is called Euler's formula.

Dot & Cross product –

Let, $z_1 = x_1 + iy_1$; $z_2 = x_2 + iy_2$

$$\therefore z_1 \cdot z_2 = x_1x_2 + y_1y_2 = |z_1||z_2|\cos\theta$$

$$z_1 \times z_2 = x_1y_2 - x_2y_1 = |z_1||z_2|\sin\theta$$

Math-

1.16

Express each of the following complex numbers in polar form-

a) $2 + 2\sqrt{3}\,i$ b) $-5 + 5i$ c) $-\sqrt{6} - \sqrt{2}\,i$ d) $-3i$

=>

a) $2 + 2\sqrt{3}\,i$

$$|r| = \sqrt{2^2 + (2\sqrt{3})^2} = 4$$

$$\theta = \tan^{-1}\left|\frac{2\sqrt{3}}{2}\right|$$

$$= \frac{\pi}{3}$$

$$\therefore 2 + 2\sqrt{3}\,i = 4\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$$

b) $-5 + 5i$

$$|r| = \sqrt{(-5)^2 + 5^2} = 5\sqrt{2}$$

$$\theta = \pi - \tan^{-1}\left|\frac{5}{5}\right|$$

$$= \pi - \frac{\pi}{4}$$

$$= \frac{3\pi}{4}$$

$$-5 + 5i = 5\sqrt{2}\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)$$

1.39

Suppose. $z_1 = 3-4i$ and $z_2 = -4+3i$. Find-

a) $z_1 \cdot z_2$

b) $|z_1 \times z_2|$

=>

a) $z_1 \cdot z_2 = Re\{3(-4) - 4(3)\} = -12 - 12 = -24$

b) $|z_1 \times z_2| = |3(3) - (-4)(-4)| = |-7| = 7$

1.40

Find the acute angle between the vectors $z_1 = 3-4i$ and $z_2 = -4+3i$.

=>

$$\cos\theta = \frac{z_1 \cdot z_2}{|z_1||z_2|} = \frac{3(-4)-4(3)}{\sqrt{3^2+4^2}\cdot\sqrt{3^2+4^2}} = \frac{-24}{25}$$

$$\therefore \theta = \cos^{-1}\left(-\frac{24}{25}\right)$$

$$= 163.74^\circ \text{ Ans}$$

c) $-6-2i$

$$|r| = \sqrt{(-6)^2+(-2)^2} = 2\sqrt{2}$$

$$\theta = \pi + \tan^{-1}\left|\frac{2}{6}\right|$$

$$= \pi + \frac{\pi}{6}$$

$$= \frac{7\pi}{6}$$

$$-6-2i = 2\sqrt{2}\left(\cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6}\right)$$

d) $-3i$

$$r = \sqrt{3^2} = 3$$

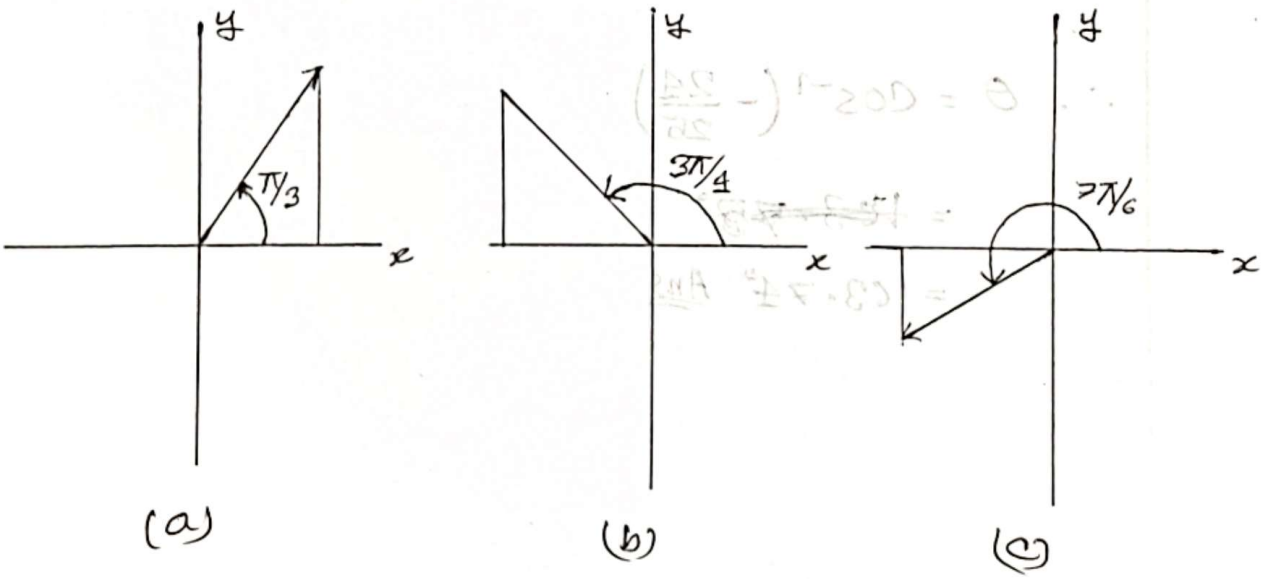
$$\theta = \pi + \tan^{-1}\left|\frac{3}{0}\right|$$

$$= \pi + \frac{\pi}{2}$$

$$= \frac{3\pi}{2}$$

$$-3i = 3\left(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}\right)$$

Graphical representation -



1.41

Prove the area of a parallelogram having sides z_1 & z_2 is $|z_1 \times z_2|$.

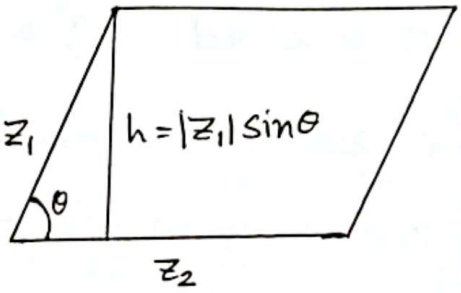
=>

Area of parallelogram

$= (|z_2|)(|z_1|\sin\theta)$

$= |z_1||z_2| \sin\theta$

$= |z_1 \times z_2|$



Vector integral

Types of integral –

1. Line integral –

Let, $A(x, y, z) = A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}$ be a vector function of position defined and continious along C, where C is assumed a finite number of curves. Then the integral of the tangential component of A along C from P_1 to P_2.

$$\int_{P_1}^{P_2} A \cdot dr = \int_C A \cdot dr = \int_C A_1 dx + A_2 dy + A_3 dz$$

Line integral $\rightarrow \int dx$

2. Surface integral –

A surface integral is an integral taken over a surface S in space.

$$\iint_S A \cdot ds = \iint_S A \cdot n ds$$

Volume integral –

Volume integral is a type of integral used to calculate values throughout a 3D region.

$$\rightarrow \iiint_V A \cdot dv$$

Definite integral –

An integral which has a finite limit that gives the exact area under a curve.

$$\rightarrow \int_a^b f(x)\, dx$$

Q-1

If $R(u) = (u-u^2)\hat{i} + 2u^3\hat{j} - 3\hat{k}$. Find-

a) $\int R(u)\,du$

b) $\int_1^2 R(u)\,du$.

—o—

a) $$\int R(u)\,du = \int (u-u^2)\hat{i}\,du + \int 2u^3\hat{j}\,du + \int -3\hat{k}\,du$$

$$= \hat{i}\left(\frac{u^2}{2} - \frac{u^3}{3}\right) + \hat{j}\left(\frac{2u^4}{4}\right) - \hat{k}(3u) + C$$

$$= \hat{i}\left(\frac{u^2}{2} - \frac{u^3}{3}\right) + \hat{j}\left(\frac{u^4}{2}\right) - \hat{k}(3u) + C$$

b) $$\int_1^2 R(u)\,du = \int_1^2 (u-u^2)\hat{i}\,du + \int_1^2 2u^3\hat{j}\,du + \int_1^2 -3\hat{k}\,du$$

$$= \hat{i}\left[\frac{u^2}{2} - \frac{u^3}{3}\right]_1^2 + \hat{j}\left[\frac{u^4}{2}\right]_1^2 - \hat{k}\left[3u\right]_1^2$$

$$= \hat{i}\left\{\frac{4}{2} - \frac{8}{3} - \frac{1}{2} + \frac{1}{3}\right\} + \hat{j}\left(\frac{32}{4} - \frac{2}{4}\right) - \hat{k}(6-3)$$

$$= -\frac{5}{6}\hat{i} + \frac{15}{2}\hat{j} - 3\hat{k}$$

Ans

Q-2

$$a = \frac{d}{dt} v = 12\cos 2t\,\hat{i} - 8\sin 2t\,\hat{j} + 16t\,\hat{k}$$

If $V = 0$ & $t = 0$; $r = 0$, $t = 0$. Find V & r.

$\Rightarrow$

$$V = \int (12\cos 2t)\,\hat{i}\,dt - \int (8\sin 2t)\,\hat{j}\,dt + \int (16t)\,\hat{k}\,dt$$

$$= 12\,\frac{\sin 2t}{2}\,\hat{i} + \frac{8\cos 2t}{2}\,\hat{j} + 16\,\frac{t^2}{2}\,\hat{k} + C_1$$

$$= (6\sin 2t)\,\hat{i} + (4\cos 2t)\,\hat{j} + (8t^2)\,\hat{k} + C_1$$

For $V = 0$, $t = 0$,

$$0 = (6\sin 2\cdot 0)\,\hat{i} + (4\cos 2\cdot 0)\,\hat{j} + (8\cdot 0^2)\,\hat{k} + C_1$$

$$\Rightarrow 0 = 4\,\hat{j} + C_1$$

$$\therefore C_1 = -4\,\hat{j}$$

$$\therefore V = 6\sin 2t\,\hat{i} + (4\cos 2t - 4)\,\hat{j} + 8t^2\,\hat{k}$$

Now,

$$r = \int 6\sin 2t\,\hat{i}\,dt + \int (4\cos 2t - 4)\,\hat{j}\,dt + \int 8t^2\,\hat{k}\,dt$$

$$= (-3\cos 2t)\,\hat{i} + (2\sin 2t - 4t)\,\hat{j} + \frac{8t^3}{3}\,\hat{k} + C_2$$

For $r=0$, $t=0$,

$$0 = -3\cos 2.0\ \hat{i} + (2\sin 2.0 - 4.0)\hat{j} + 0\hat{k} + c_2$$

$$\Rightarrow c_2 = 3\hat{i}$$

$$\therefore r = (3 - 3\cos 2t)\hat{i} + (2\sin 2t - 4t)\hat{j} + \frac{8}{3}t^3\hat{k}$$

Q-6

If $A = (3x^2+6y)\hat{i} - 14yz\,\hat{j} + 20xz^2\,\hat{k}$, evaluate $\int_C A \cdot dr$ from $(0,0,0)$ to $(1,1,1)$.

a) $x = t$, $y = t^2$, $z = t^3$

b) the straight line from $(0,0,0)$ to $(1,0,0)$ then to $(1,1,0)$ and then to $(1,1,1)$.

c) the straight line joining $(0,0,0)$ and $(1,1,1)$.

$\Rightarrow$

$$\int_C A \cdot dr = \int_C [(3x^2+6y)\hat{i} - 14yz\,\hat{j} + 20xz^2\,\hat{k}] \cdot (dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k}).$$

$$= \int_C (3x^2+6y)\,dx + -14yz\,dy + 20xz^2\,dz$$

a) $x = t$, $y = t^2$, $z = t^3$ points $(0,0,0)$ and $(1,1,1)$ correspond to $t = 0$, $t = 1$ respectively.

$$\int_C A \, dr = \int_{t=0}^{1} (3t^2 + 6t^2) \, dt$$

$$-14 (t^2)(t^3) 2t \cdot dt$$

$$+ 20t (t^3)^2 \cdot 3t^2 \, dt$$

$$= \int_{t=0}^{1} 9t^2 \, dt - \int_{t=0}^{1} 28 t^6 \, dt + \int_{t=0}^{1} 60 t^9 \, dt$$

$$= \left[\frac{9t^3}{3}\right]_0^1 - \left[\frac{28t^7}{7}\right]_0^1 + \left[60 \frac{t^{10}}{10}\right]_0^1$$

$$= (3 - 4 + 6)$$

$$= 5$$

Ans

$x = t$

$\Rightarrow \frac{d}{dt} x = \frac{d}{dt} t$

$\therefore dx = dt$

$y = t^2$

$\Rightarrow \frac{d}{dt} y = \frac{d}{dt} t^2 = 2t$

$\Rightarrow dy = 2t \, dt$

$z = t^3$

$\Rightarrow dz = 3t^2 \, dt$

b) From $(0,0,0)$ to $(1,0,0)$ – $y=0$, $z=0$; $dy=0$, $dz=0$.

$$\int_{x=0}^{1} \{3x^2 + 6\cdot 0\} dx - 0 + 0$$

$$= [x^3]_0^1$$

$$= 1$$

From $(1,0,0)$ to $(1,1,0)$ – $x=1$, $z=0$; $dx=0$, $dz=0$

$$\int_{y=0}^{1} -14\cdot y\cdot 0 \cdot dy + 0$$

$$= 0$$

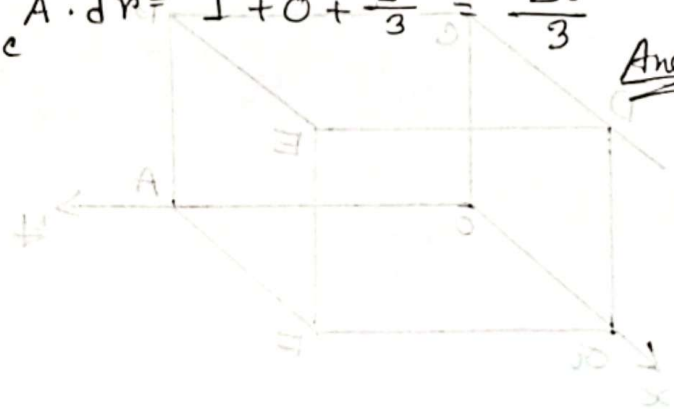
From $(1,1,0)$ to $(1,1,1)$ – $x=1$, $y=1$, $dx=0$, $dy=0$

$$\int_{z=0}^{1} (20)(1)\, z^2\, dz$$

$$= \left[\frac{20z^3}{3}\right]_0^1$$

$$= \frac{20}{3}$$

$$\therefore \int_c A \cdot dr = 1 + 0 + \frac{20}{3} = \frac{23}{3}$$ Ans



(c) the straight line joining (0,0,0) and (1,1,1) is given parameric from $x=t$, $y=t$, $z=t$.

$$\int_C A \cdot dr = \int_{t=0}^{1} (3t^2+ct)\,dt - 14\,t\cdot t\cdot dt + 20\cdot t\cdot t^2\,dt$$

$$= \int_0^1 (3t^2+ct-14t^2+20t^3)\,dt$$

$$= \left[3t^2 - \frac{11t^3}{3} + 5t^4\right]_0^1$$

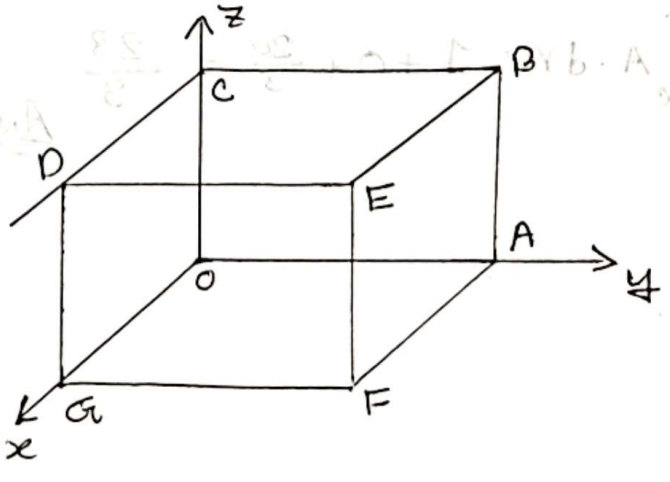
$$= 3 - \frac{11}{3} + 5$$

$$= \frac{13}{3}$$

Ans

Q-23

If $F = 4xz\,\hat{i} - y^2\,\hat{j} + yz\,\hat{k}$: Evaluate $\iint F \cdot n\,ds$ where S is the surface of the cube bounded by $x=0$, $x=1$, $y=0$, $y=1$, $z=0$, $z=1$.



Face DEFG : $n = \hat{i}$, $x = 1$.

$$\iint_{DEFG} F \cdot n \, ds = \int_0^1 \int_0^1 (4 \cdot 1 \cdot z \, \hat{i} - y^2 \hat{j} + yz \, \hat{k}) \cdot \hat{i} \, dy \, dz$$

$$= \int_0^1 \int_0^1 4z \, dz \cdot dy$$

$$= \int_0^1 [2z^2]_0^1 \, dy$$

$$= \int_0^1 2 \, dy$$

$$= [2y]_0^1$$

$$= 2$$

Face ABCO : $n = -\hat{i}$, $x = 0$

$$\iint_{ABCO} F \cdot n \, ds = \int_0^1 \int_0^1 (-y^2 \hat{j} + yz \, \hat{k}) \cdot (-\hat{i}) \, dy \, dz$$

$$= 0$$

Face ABEF : $n = \hat{j}$, $y = 1$

$$\iint_{ABEF} F \cdot n \, ds = \int_0^1 \int_0^1 (4xz \, \hat{i} - \hat{j} + z \, \hat{k}) \cdot \hat{j} \, dx \, dz$$

$$= \int_0^1 \int_0^1 [-1] \, dx \, dz$$

$$= \int_0^1 [-x]_0^1 \, dz = \int_0^1 -1 \, dz = [-1]_0^1 = -1$$

Face OGDC : $n = -\hat{j}$, $y = 0$

$$\iint_{OGDC} F.n\,ds = \int_0^1 \int_0^1 (4xz\,\hat{i} - 0 + 0)\,(-\hat{j})\,dx\,dz$$

$$= 0$$

Face BCDE : $n = \hat{k}$, $z = 1$

$$\iint_{BCDE} F.n\,ds = \int_0^1 \int_0^1 (4x\,\hat{i} - y^2\,\hat{j} + y\,\hat{k}).\hat{k}\,dx\,dy$$

$$= \int_0^1 \int_0^1 y\,dx\,dy$$

$$= \int_0^1 \left[\frac{y^2}{2}\right]_0^1 dx$$

$$= \int_0^1 \frac{1}{2}\,dx$$

$$= \left[\frac{x}{2}\right]_0^1 = \frac{1}{2}$$

Face AFGO : $n = -\hat{k}$, $z = 0$

$$\iint_{AFGO} F.n\,ds = \int_0^1 \int_0^1 (-y^2\,\hat{j})\,(-\hat{k})\,dx\,dy$$

$$= 0$$

$$\therefore \iint F.n\,ds = 2 + 0 - 1 + 0 + \frac{1}{2} + 0 = \frac{3}{2}$$ Ans

Chp-13 Fourier series

Even signal ->

An even signal is a signal that's symmetric about the vertical.

Even signal satisfy this condition, $x(t) = x(-t)$.

Cos -wave is always even.

Odd signal ->

An odd signal is a signal that's symmetric about the origin.

Odd signal satisfy this condition. $x(t) = -x(-t)$.

Sin - wave is always odd.

Sinusoidal signal ->

A sinusoidal signal is a signal that follows the shape of a sine or cosine wave.

General equation -> $x(t) = A\sin(\omega t + \theta) = A\cos(\omega t + \theta)$

Periodic signal ->

A signal that repeats after a fixed interval or time period T, is called periodic signal.

Fourier series -

A Fourier series is a method to represent any periodic signal as a sum of simple sine and cosine waves.

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}$$

Here,

$$a_0 = \frac{1}{L} \int_{-L}^{L} f(x)\, dx$$

$$a_n = \frac{1}{L} \int_{-L}^{L} f(x) \cos \frac{n\pi x}{L}\, dx$$

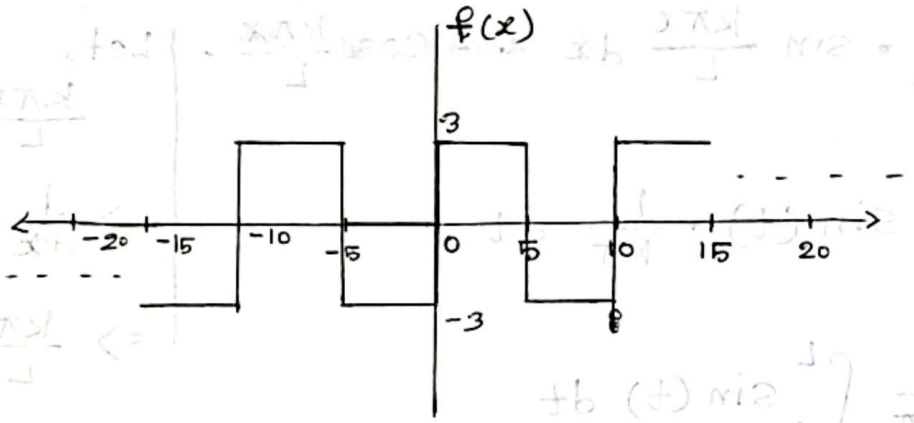
$$b_n = \frac{1}{L} \int_{-L}^{L} f(x) \sin \frac{n\pi x}{L}\, dx$$

Necessity of Fourier series -

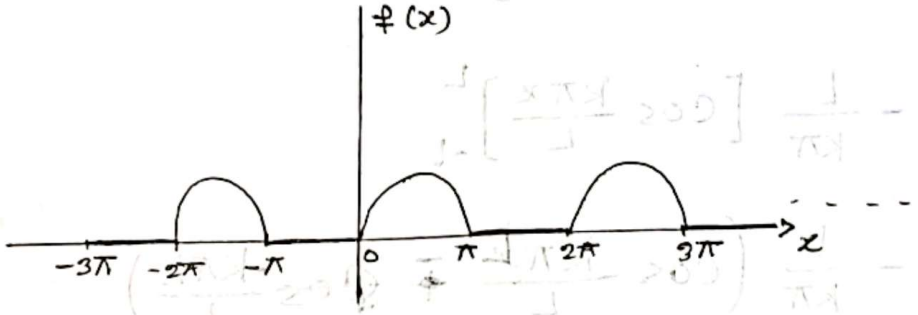
1. Simplifies complex signals.
2. Easy analysis of systems.
3. Helps to analyze AC circuit.
4. Useful for removing unwanted frequencies from signals.
5. Useful for signal processing.
6. Help to solve differential wave equations.

Ex - 13.1

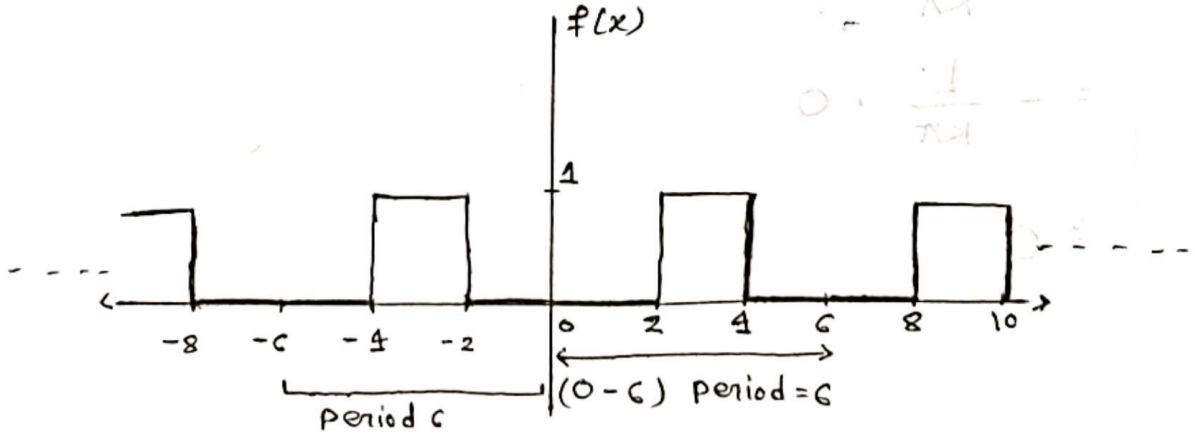
a) $f(x) = \begin{cases} 3 & 0 < x < 5 \\ -3 & -5 < x < 0 \end{cases}$ period = 10



b) $f(x) = \begin{cases} \sin x & 0 \le x \le \pi \\ 0 & \pi < x < 2\pi \end{cases}$ period = 2π



c) $f(x) = \begin{cases} 0 & 0 \le x < 2 \\ 1 & 2 \le x < 4 \\ 0 & 4 \le x < 6 \end{cases}$ period = 6



13.2

Prove $\int_{-L}^{L} \sin\frac{k\pi x}{L}\,dx = \int_{-L}^{L} \cos\frac{k\pi x}{L}\,dx = 0$; if $k = 0, 1, 2 \cdots$

$\Rightarrow$

$$\int_{-L}^{L} \sin\frac{k\pi x}{L}\,dx$$

Let,

$$\frac{k\pi x}{L} = t$$

$$\Rightarrow \frac{d}{dx}\frac{k\pi x}{L} = \frac{dt}{dx}$$

$$\Rightarrow \frac{k\pi}{L} = \frac{dt}{dx}$$

$$\therefore dx = \frac{L}{k\pi}\,dt$$

$$= \int_{-L}^{L} \sin(t) \cdot \frac{L}{k\pi}\,dt$$

$$= \frac{L}{k\pi}\int_{-L}^{L} \sin(t)\,dt$$

$$= -\frac{L}{k\pi}\left[\cos(t)\right]_{-L}^{L}$$

$$= -\frac{L}{k\pi}\left[\cos\frac{k\pi x}{L}\right]_{-L}^{L}$$

$$= -\frac{L}{k\pi}\left(\cos\frac{k\pi k}{L} - \cos\frac{-k\pi L}{L}\right)$$

$$= -\frac{L}{k\pi}\left(\cos k\pi - \cos(-k\pi)\right)$$

$$= -\frac{L}{k\pi}\left(\cos k\pi - \cos k\pi\right)$$

$$= -\frac{L}{k\pi} \cdot 0$$

$$= 0$$

Now,

$$\int_{-L}^{L} \cos \frac{k\pi x}{L}\, dx$$

Let,

$$\frac{k\pi x}{L} = t$$

$$\Rightarrow \frac{d}{dx}\frac{k\pi x}{L} = \frac{dt}{dx}$$

$$\therefore dx = \frac{L}{k\pi}\, dt$$

$$= \int_{-L}^{L} \cos(t) \cdot \frac{L}{k\pi}\, dt$$

$$= \frac{L}{k\pi}\left[\sin(t)\right]_{-L}^{L}$$

$$= \frac{L}{k\pi}\left[\sin \frac{k\pi x}{L}\right]_{-L}^{L}$$

$$= \frac{L}{k\pi}\left(\sin k\pi - \sin(-k\pi)\right)$$

$$= \frac{L}{k\pi}\sin k\pi + \frac{L}{k\pi}\sin k\pi$$

$$= 0 + 0$$

$$= 0$$

$(k = 1, 2, 3, \ldots)$

$(\sin \pi = 0,\ \sin 2\pi = 0)$

Ex-13.5

$$\begin{cases} 0 & -5 < x < 0 \\ 3 & 0 < x < 5 \end{cases} \qquad \text{period} = 10.$$

a) Find Fourier series.

=>

a) Period $= 2L = 10 \quad \therefore L = 5$

Now,

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}\,dx.$$

$$= \frac{1}{5}\int_{-5}^{5} f(x)\cos\frac{n\pi x}{5}\,dx$$

$$= \frac{1}{5}\int_{-5}^{0} f(x)\cos\frac{n\pi x}{5}\,dx + \frac{1}{5}\int_{0}^{5} f(x)\cos\frac{n\pi x}{5}\,dx$$

$$= \frac{1}{5}\int_{-5}^{0} 0\cdot\cos\frac{n\pi x}{5}\,dx + \frac{1}{5}\int_{0}^{5} 3\cos\frac{n\pi x}{5}\,dx$$

$$= 0 + \frac{1}{5}\int_{0}^{5} 3\cos\frac{n\pi x}{5}\,dx$$

$$= \frac{3}{5}\left[\sin\frac{n\pi x}{5}\cdot\frac{n\pi}{5}\,\frac{x^2}{2}\right]_0^5$$

$$= \frac{3}{5}\cdot\frac{n\pi}{5}\left[\frac{x^2}{2}\sin\frac{n\pi x}{5}\right]_0^5$$

$$= \frac{3}{5} \cdot \frac{n\pi}{5} \left[\frac{5^2}{2} \sin \frac{5 n\pi}{5} \right] - 0 \Big]$$

$$= \frac{3}{5} \frac{n\pi}{5} \left[0 - 0 \right] \qquad \rightarrow \text{As } \sin(n\pi) = 0$$

$$= 0$$

$$b_n = \frac{1}{L} \int_{-L}^{L} f(x) \sin \frac{n\pi x}{L} \, dx$$

$$= \frac{1}{5} \int_{-5}^{5} f(x) \sin \frac{n\pi x}{5} \, dx$$

$$= \frac{1}{5} \int_{0}^{5} 3 \sin \frac{n\pi x}{5} \, dx$$

$$= \frac{3}{5} \left[-\cos \frac{n\pi x}{5} \cdot \frac{n\pi}{5} \frac{x^2}{2} \right]_0^5$$

$$= \frac{3}{5} \frac{n\pi}{5} \left[-\cos \frac{5 n\pi}{5} \cdot \frac{5^2}{2} - 0 \right]$$

$$= -\frac{3n\pi}{25} \cdot \frac{25}{2} (\cos n\pi)$$

$$= -\frac{3n\pi}{2} \cos n\pi$$

$$= -\frac{3n\pi}{2} (-1)^n \qquad \rightarrow \cos(n\pi) = (-1)^n$$

Now,

$$a_0 = \frac{1}{L}\int_{-L}^{L} f(x)\,dx$$

$$= \frac{1}{5}\int_{-5}^{5} f(x)\,dx$$

$$= \frac{1}{5}\int_{0}^{5} 3\,dx$$

$$= \frac{1}{5}\,[3x]_0^5$$

$$= \frac{1}{5}\cdot 15$$

$$= 3$$

Fourier series,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cdot \cos\frac{n\pi x}{L} + \sum_{n=1}^{\infty} b_n \cdot \sin\frac{n\pi x}{L}$$

$$= \frac{3}{2} + \sum_{n=1}^{\infty} 0 \cdot \cos\frac{n\pi x}{L} + \sum_{n=1}^{\infty} -\frac{3\pi n}{2}(-1)^n \cdot \sin\frac{n\pi x}{L}$$

$$= \frac{3}{2} + \sum_{n=1}^{\infty} \left(-\frac{3\pi n}{2}(-1)^n \sin\frac{n\pi x}{L}\right)$$

If,

$$n = 1, \quad \frac{-3\pi}{2}(-1)^1 \sin\frac{\pi x}{5} = \frac{3\pi}{2}\sin\frac{\pi x}{5}$$

$$n = 2, \quad \frac{-3\pi 2}{2}(-1)^2 \sin\frac{2\pi x}{5} = -3\pi \sin\frac{2\pi x}{5}$$

$$n = 3, \quad \frac{-3\pi 3}{2}(-1)^3 \sin\frac{3\pi x}{5} = -\frac{9\pi}{2}\sin\frac{3\pi x}{5}$$

$$\therefore f(x) = \frac{3}{2} + \frac{3\pi}{2}\sin\frac{\pi x}{5} - 3\pi \sin\frac{2\pi x}{5} + \frac{9\pi}{2}\sin\frac{3\pi x}{5}$$

Ans